

Rehearsal

An AI-assisted study planner for Pakistani Matric students — AI App Development Competition Submission

Topic	Study Planner for Matric Students
Team / Participant name(s)	[Add your name(s) here]
App link	[Add your deployed URL here, e.g. https://your-app.netlify.app]
Demo video	[Add your video link here]
Repository	[Add your GitHub link here, optional]

1. The Problem

Matric (Class IX–X) students in Pakistan prepare for high-stakes board exams that determine their group placement in college — yet most students plan their revision on scattered paper timetables, WhatsApp reminders, or not at all. This causes three recurring problems:

- No group-specific structure — a Pre-Medical student and a Computer Science student have almost entirely different subjects, but generic "study apps" treat every student the same.
- No visibility into progress — students can't see, at a glance, how much of a subject's syllabus is actually covered versus how much time is left before the board exam.
- No affordable, on-demand help — many students can't access a tutor for a quick explanation or a practice quiz at 11 p.m. the night before a test.

Rehearsal solves this by combining a group-aware syllabus tracker, a weekly study planner, and an AI study coach into a single lightweight, installable mobile app — free to build, free to host, and free for students to use.

2. Target Users

Matric students (Class IX–X, roughly ages 14–16) across Pakistani boards (BISE Lahore, Karachi, Federal Board, and others) in five common subject groups:

- Science – Pre-Engineering (Physics, Chemistry, Mathematics)
- Science – Pre-Medical (Physics, Chemistry, Biology)
- Computer Science (Computer Science, Physics, Mathematics)
- Humanities / Arts (Economics, Civics, General Math)
- General Science / Home Economics

3. What the App Does

- **Onboarding** — student enters their name, exam date, and matric group, and picks a starting visual theme.
- **Subjects** — each group's real subjects and chapter lists are pre-loaded; students tap chapters off as they revise them, and see a live completion percentage per subject.
- **Planner** — a weekly timetable for scheduling study sessions by subject, day and time, plus a simple to-do list for tasks like "solve Physics Ch. 3 past paper."
- **AI Coach** — a chat-based tutor (see Section 4) that can build a week's study plan, explain a topic in simple language, quiz the student one question at a time, or suggest ways to beat procrastination.
- **Progress** — a streak counter, per-subject progress bars, and unlockable badges (first topic completed, 3-day streak, subject mastered, etc.) to keep students motivated.
- **Settings** — switch between five distinct visual themes, edit the profile/group, manage the AI key, export a JSON backup, or reset the app.

4. AI Tools Used

The in-app **AI Coach** is powered by the **Groq API** (OpenAI-compatible chat completion endpoint), using the **llama-3.3-70b-versatile** model. The app calls Groq directly from the browser using a student-supplied, free Groq API key — there is no backend server, so the key is stored only in the browser's local storage and is sent only to Groq's endpoint.

The AI is used for:

- Generating a personalized weekly study plan from the student's matric group and subjects.
- Explaining syllabus topics in simple, exam-focused language with examples.
- Running one-question-at-a-time quizzes on a topic the student chooses.
- Practical, non-judgmental advice for procrastination and exam anxiety.

A dedicated system prompt (see `js/data.js`, `AI_SYSTEM_PROMPT`) grounds every response in the Pakistani matric context — Pre-Engineering, Pre-Medical, Computer Science, Humanities, and General Science/Home Economics groups, and the boards that examine them — so answers stay relevant and age-appropriate.

Development itself was also AI-assisted: the app's structure, styling system, and content (including the matric subject/chapter lists) were drafted with the help of an AI coding assistant and then reviewed and adapted for this submission.

5. Technology & Architecture

- Vanilla HTML, CSS and JavaScript — no framework or build step, so it is easy to read, modify, and host anywhere.
- A CSS custom-property theme system powers five switchable visual themes without duplicating markup.
- All student data (profile, syllabus progress, planner, chat history, badges) is stored client-side in `localStorage` — no database or backend needed.

- A web app manifest and service worker make the app installable to a phone's home screen and usable offline (except for live AI Coach requests, which need a connection).

6. How the App Meets the Judging Criteria

Criterion	How Rehearsal addresses it
Problem Impact	Targets a real, daily pain point for every matric student: fragmented, generic exam preparation.
Functionality	Full loop implemented: onboarding, syllabus tracking, planner, AI coach, progress, settings — all working.
User Interface (UI/UX)	Mobile-first "roll-number slip" visual identity, five complete themes, bottom tab navigation, smooth transitions.
Demo	Guided flow from onboarding to AI chat to progress screen, ready for a 2–3 minute walkthrough.
AI Integration	Live Groq-powered chat coach with syllabus-aware system prompt and quick-action prompts.
Practical Usefulness	Free to host, free to use, works on any phone browser, installable, and usable fully offline apart from live AI requests.

Full source code, setup instructions, and deployment steps are included in the project's README.md.